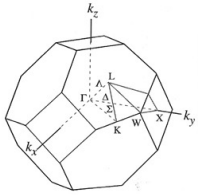
Carrier Transport

Baile Chen

School of Information Science and Technology

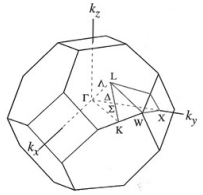
Shanghai Tech University

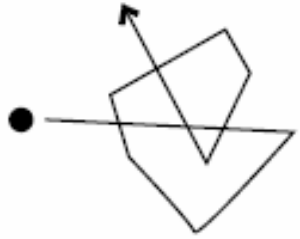


What We Learned So Far?

- We know how to calculate the carrier density.
- What is the next step? Calculate the velocity.
  - ➢ The secret of C-band electrons (or V-band holes): They are essentially free to move around at finite temperature & doping. So what do they do?
  - ➢ Instantaneous velocity given by thermal energy (3/2kT).
    - ✓ Eg: T=300K, electron effective mass in GaAs: 0.067m*
    - ✓ Around 4.3×10^7 cm/s
  - ➢ Scattering time: average time between collisions is of the order ~0.1- 1ps.
    - ✓ 0.1ps for GaAs
  - ➢ So average distance travelled between scattering: L

$$L = v_{th} \cdot \tau$$





Turn ON an Electric Field

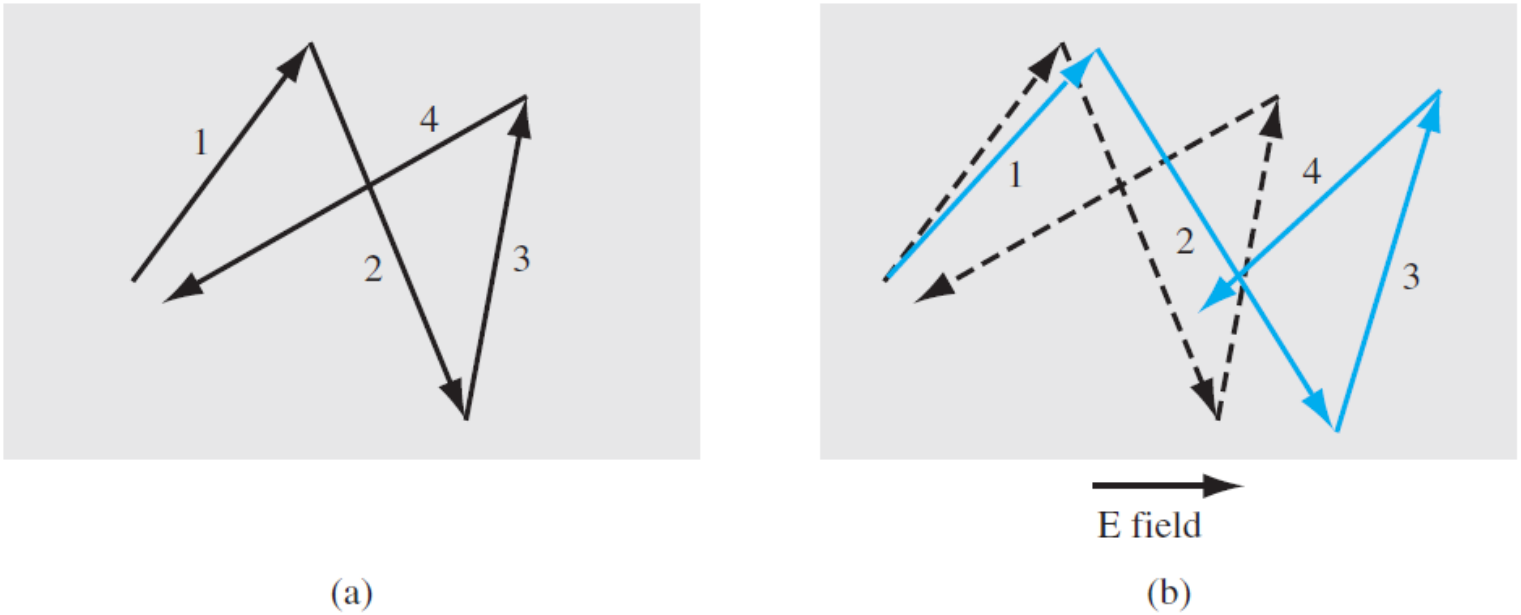
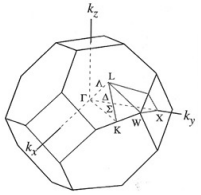
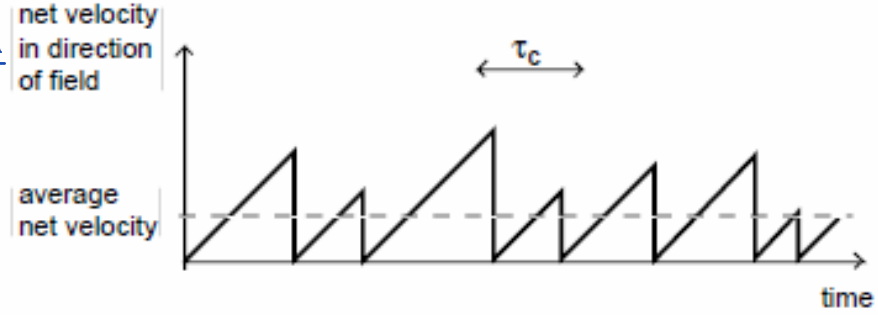


Figure 5.1 | Typical random behavior of a hole in a semiconductor (a) without an electric field and (b) with an electric field.

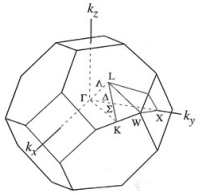


Turn ON an Electric Field

- For electrons: $F = m_n^* a_n = -eE$
  - $v_n = \dfrac{-eE\tau_n}{m_n^*}$ $\quad a_n = \dfrac{-eE}{m_n^*}$, $v_n = \dfrac{-eE\tau_n}{m_n^*}$
- For holes: $F = m_p^* a_p = eE$
  - $v_p = \dfrac{eE\tau_p}{m_p^*}$



- Because of Scattering, electrons/holes in a semiconductor do not achieve constant acceleration. However, they can be viewed as quasi-classical particles moving at a constant average drift velocity

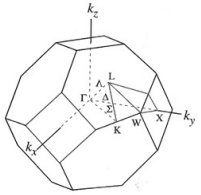


Mobility Effects

- For electron: $F = m_n^* a_n = -eE$
  - $v_n = \frac{-eE\tau_n}{m_n^*}$
  - $\mu_n = \frac{e\tau_n}{m_n^*}$
- For hole: $F = m_p^* a_p = eE$
  - $v_p = \frac{eE\tau_p}{m_p^*}$
  - $\mu_p = \frac{e\tau_p}{m_p^*}$
- What are the roles of $m_{n/p}^*$ and $\boldsymbol{\tau}_{n/p}$
- Mobility is a measure of *ease of carrier drift in E-field.*
  - If $m\downarrow$ "lighter" particle means μ…
  - If $\boldsymbol{\tau}\uparrow$ means longer time between collisions, so μ…
  - Mobility has the dimensions of v/E: cm/s/V/cm = $cm^2/V/s$

Electron and hole mobilities of selected intrinsic semiconductors (*T*=300K)

	Si	Ge	GaAs	InAs
μ_n ($cm^2/V{\cdot}s$)	1400	3900	8500	30000
μ_p ($cm^2/V{\cdot}s$)	470	1900	400	500

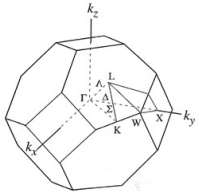


Carrier Drift

- Drift Current Density
  - Definition: the number of charges passed unit area in unit of time
  - $J_{dft} = \rho v$ with unit of A/cm^2.
- For electrons:
  - $J_{n|dft} = -env_{dn}$
  - $v_{dn} = -\mu_n E$
  - $J_{n|dft} = en\mu_n E$
- For holes:
  - $J_{p|dft} = epv_{dp}$
  - $v_{dp} = \mu_p E$
  - $J_{p|dft} = ep\mu_p E$

$$J_{dft} = J_{n|dft} + J_{p|dft} = en\mu_n E + ep\mu_p E = \sigma E$$

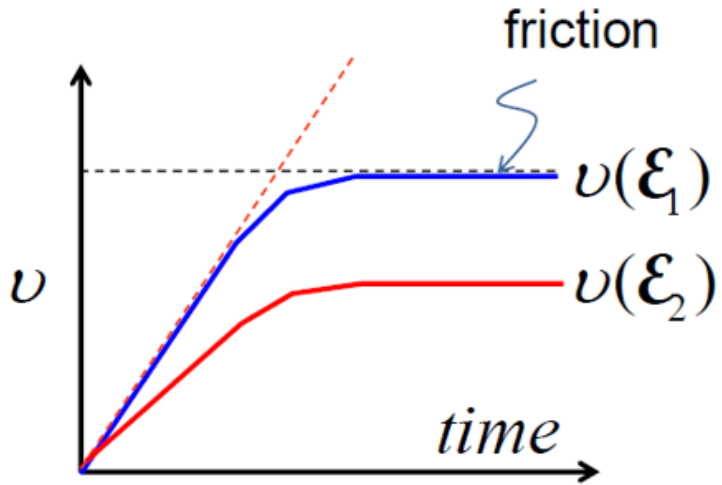
$$\sigma = e(n\mu_n + p\mu_p) = \frac{1}{\rho}$$

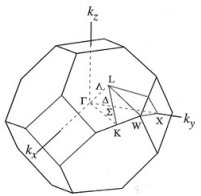


Mobility Effects

- Why does a Field Create a Velocity rather than an Acceleration?
  - $m_n^*(dv/dt + v/\tau_n) = -qE$
  - $v(t) = -\frac{q\tau_n}{m_n^*}E\left(1 - e^{-\frac{t}{\tau_n}}\right) = -\frac{q\tau_n}{m_n^*}E = -\mu_n E$ *when* $t \to \infty$, *(1-2ps)*
  - $\mu_n = q\tau_n/m_n^*$

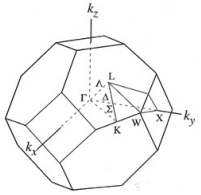
Theory valid once t>1-2 ps





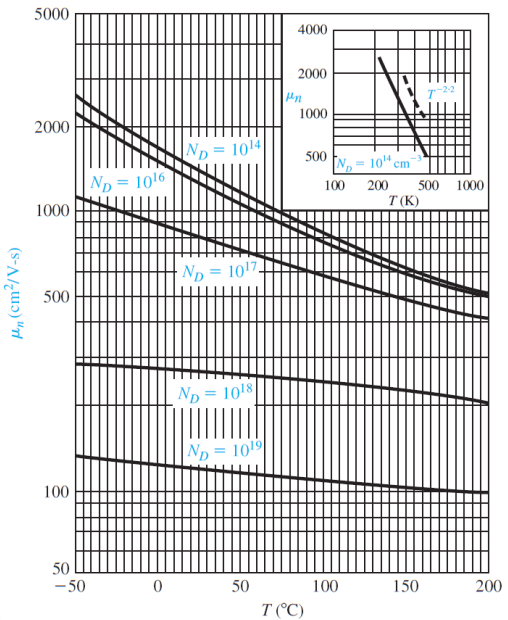
What Causes Scattering?

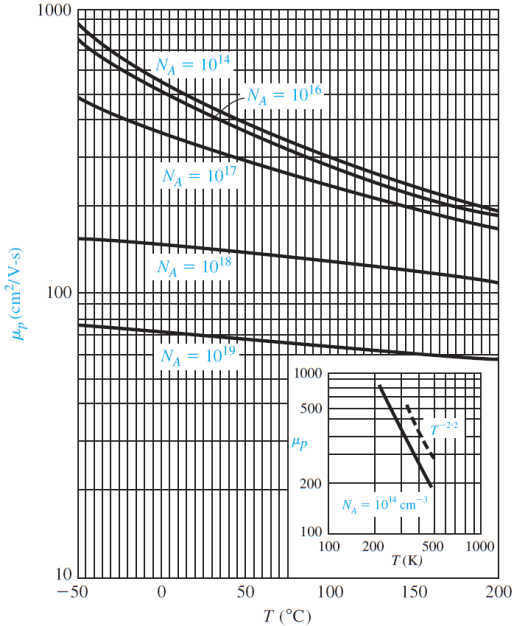
- Lattice Scattering (Phonon Scattering)
- Ionized Impurity Scattering
- Neutral Atom/Defect Scattering
- Carrier-Carrier Scattering
- Piezoelectric Scattering

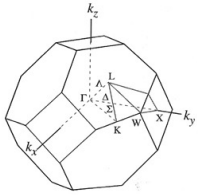


Lattice Scattering

- $\mu_L \propto T^{-\frac{3}{2}}$ $T\uparrow \cdot \frac{1}{\tau_L}\uparrow \quad \mu_L\downarrow$
- Lattice vibrations decrease as the temperature decreases







Ionized Impurity Scattering

- $\mu_I \propto \frac{T^{\frac{3}{2}}}{N_I}$
- $N_I = N_d^+ + N_a^-$

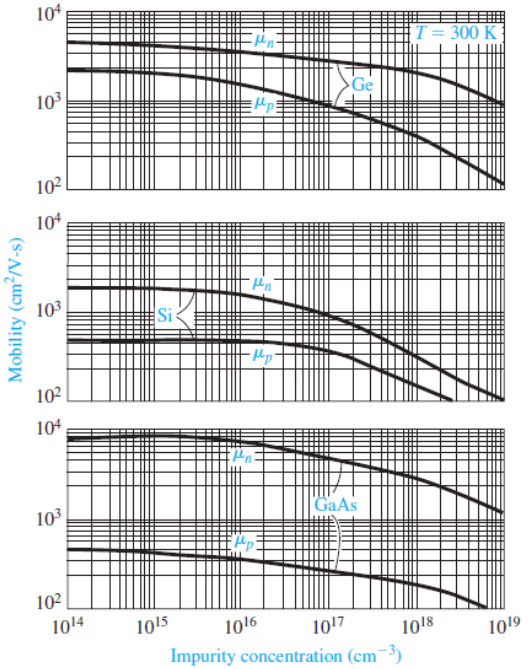
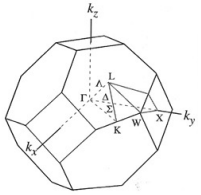
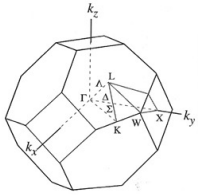


Figure 5.3 | Electron and hole mobilities versus impurity concentrations for germanium, silicon, and gallium arsenide at $T = 300$ K.
(From Sze [14].)

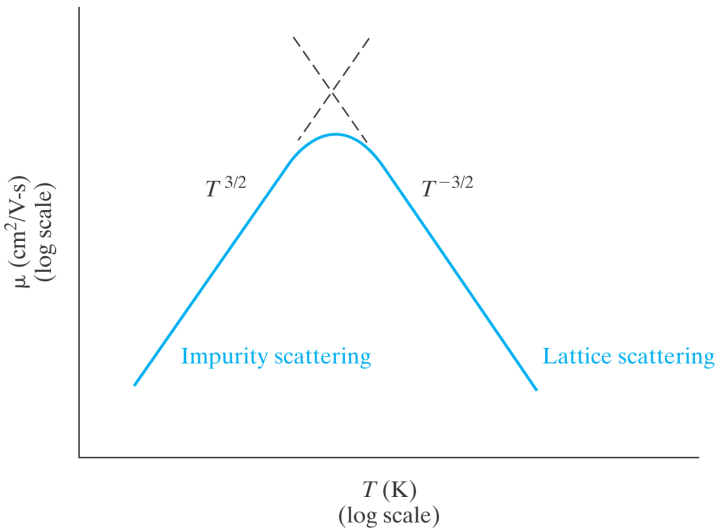


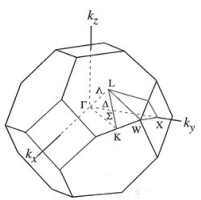
Multiple Scattering Processes

- The total probability of a scattering event occurring in the differential time dt:
  - $\frac{dt}{\tau} = \frac{dt}{\tau_1} + \frac{dt}{\tau_2} + \cdots$
  - $\frac{1}{\mu} \propto \frac{1}{\tau}$
- The net mobility follows:
  - $\frac{1}{\mu} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \cdots$



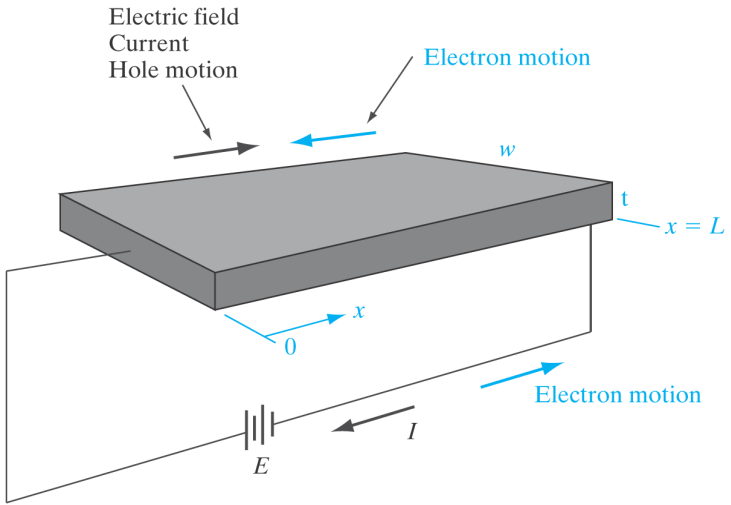
Qualitatively

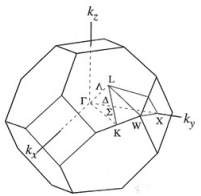




Conductivity

- $J_{dft} = e(\mu_n n + \mu_p p)E = \sigma E$
- Resistivity: $\rho = \frac{1}{\sigma} = \frac{1}{e(\mu_n n + \mu_p p)}$
- Conductivity $\sigma = en\mu_n + ep\mu_p$
- Ohm's Law
  - $\frac{I}{A} = \sigma\left(\frac{V}{L}\right)$
  - $I = V\left(\frac{A\sigma}{L}\right) = \frac{V}{R}$
  - $R = \frac{L}{A\sigma}$

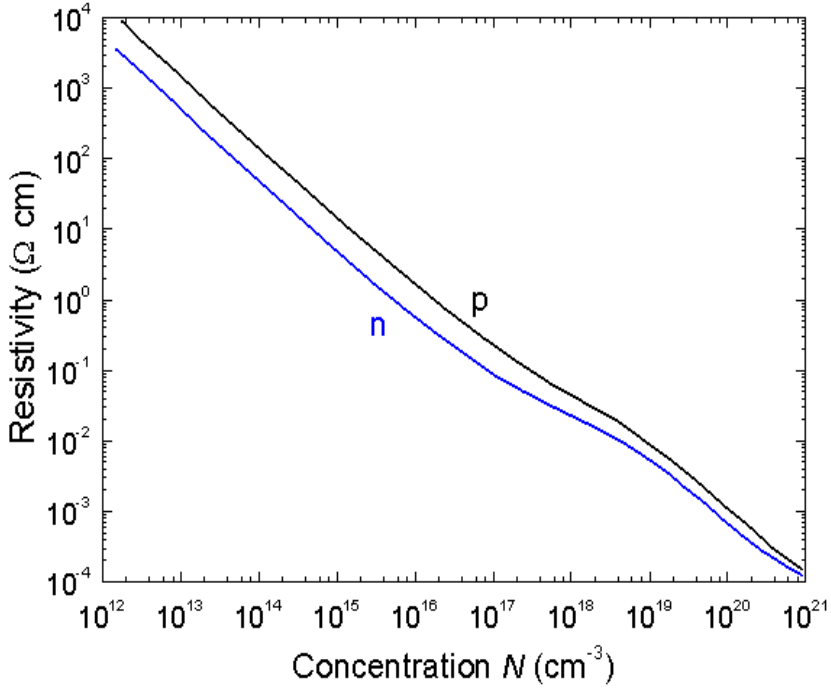


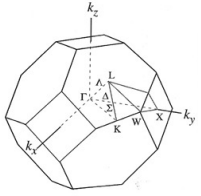


Experimentally, for Silicon at Room Temperature:

- This is absolutely essential to illustrating our control over a semiconductor's resistivity via doping.

- Notes:
  - This plot does not apply to compensated materia (with <u>both</u> n-type and p-type dopant impurities).
  - This applies at relatively low fields (at high field we must be a little careful with velocity saturation).





Velocity Saturation in Si/Ge

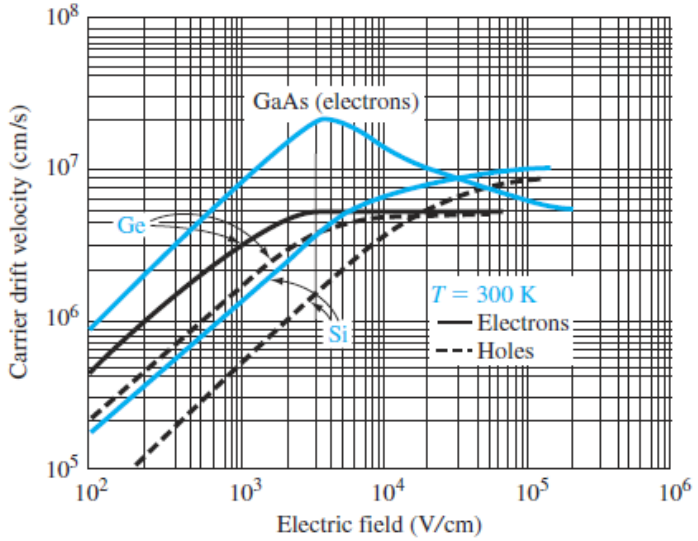
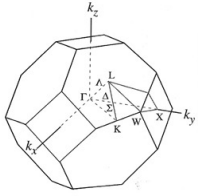
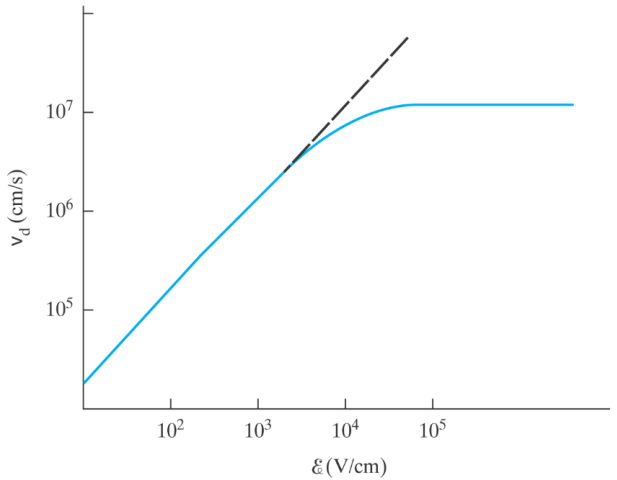


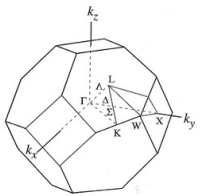
Figure 5.7 | Carrier drift velocity versus electric field for high-purity silicon, germanium, and gallium arsenide. (*From Sze [14].*)



Electrons in Silicon

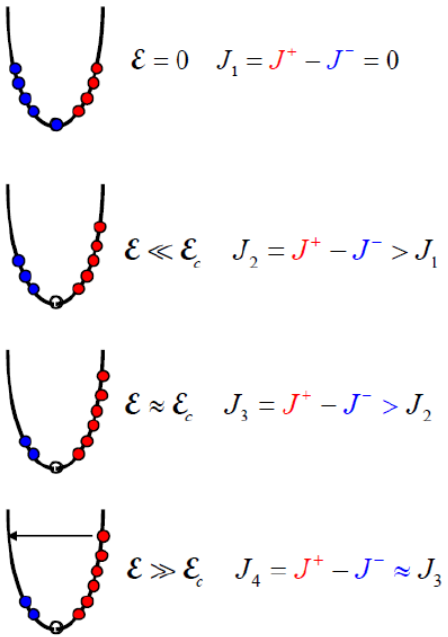
- Low-field slope = mobility
- High-field effect = velocity saturation due to very strong lattice scattering. Any additional energy from the E-field is transferred to the lattice (phonons) rather than increasing the carrier velocity.
- Result: constant velocity (current) at very high fields!

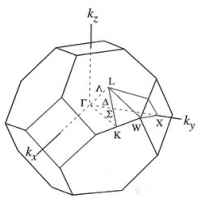




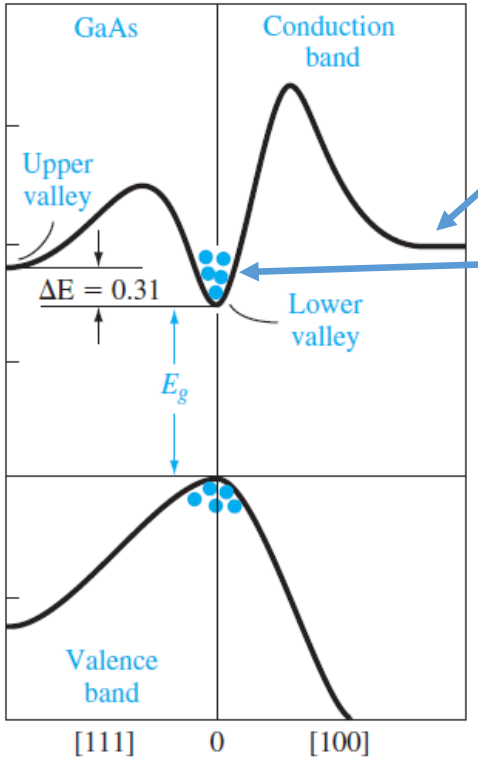
Velocity versus Electric Field for Silicon

- $v_n = \dfrac{v_s}{\left[1+\left|\dfrac{E_{on}}{E}\right|^2\right]^{1/2}}$
- $v_p = \dfrac{v_s}{\left[1+\left|\dfrac{E_{op}}{E}\right|\right]}$



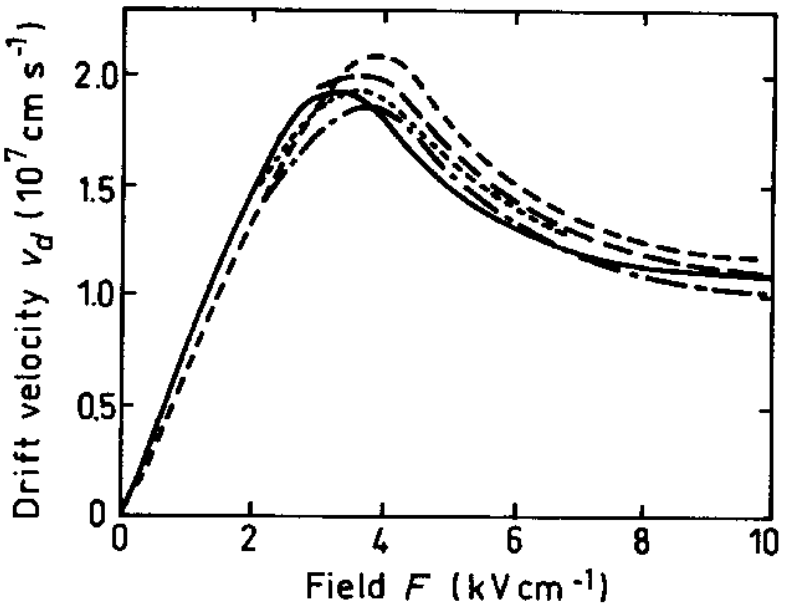


Velocity Overshoot for GaAs

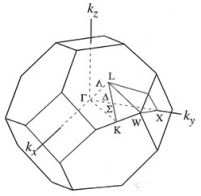


Larger effective mass

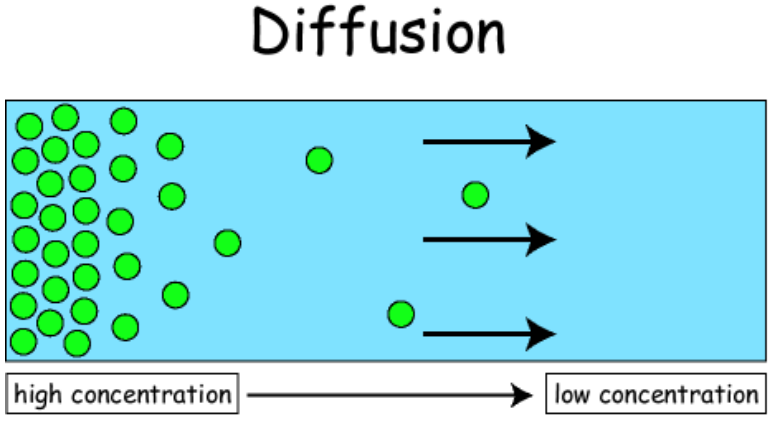
Smaller effective mass



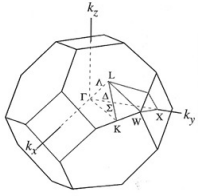
•Blakemore, J. S., *J. Appl. Phys.* **53**, 10 (1982) R123-R181.



Carrier Diffusion



Solute transport is from the left to the right; movement of the solutes is due to the concentration gradient (dC/dx).



Carrier Diffusion

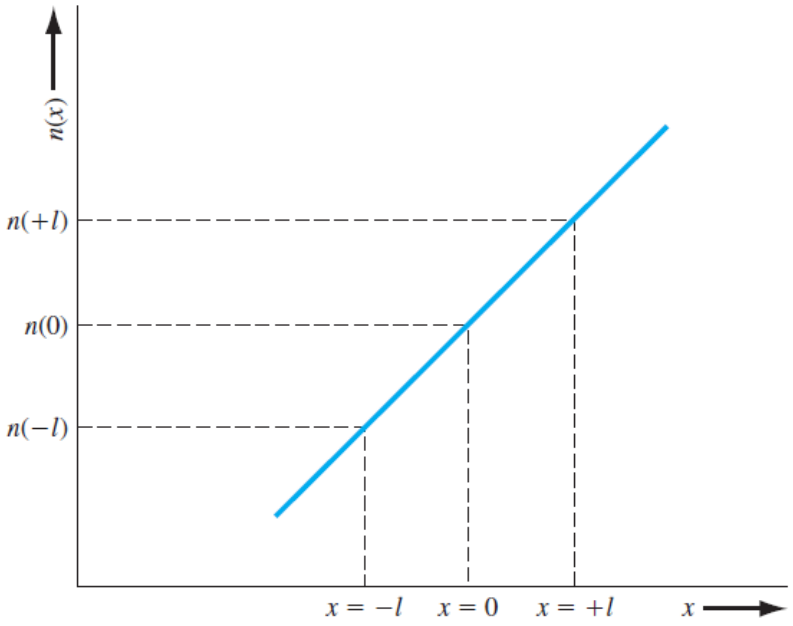
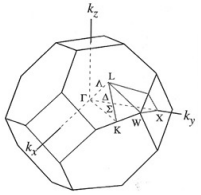


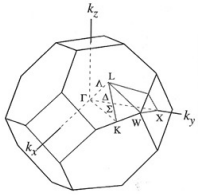
Figure 5.10 | Electron concentration versus distance.

- Electron flow:
  - $F_n = \frac{1}{2}n(-l/2)v_{th} - \frac{1}{2}n(+l/2)v_{th}$
  - $F_n = -v_{th}l/2\frac{dn}{dx}$
  - l is the mean free path
- Electron Current:
  - $J_{n|diff} = -eF_n = ev_{th}l/2\frac{dn}{dx}$
  - $J_{n|diff} = eD_n\frac{dn}{dx}$
  - $D_n \propto \mu_n$
- Similar for hole
  - $J_{p|diff} = -eD_p\frac{dp}{dx}$

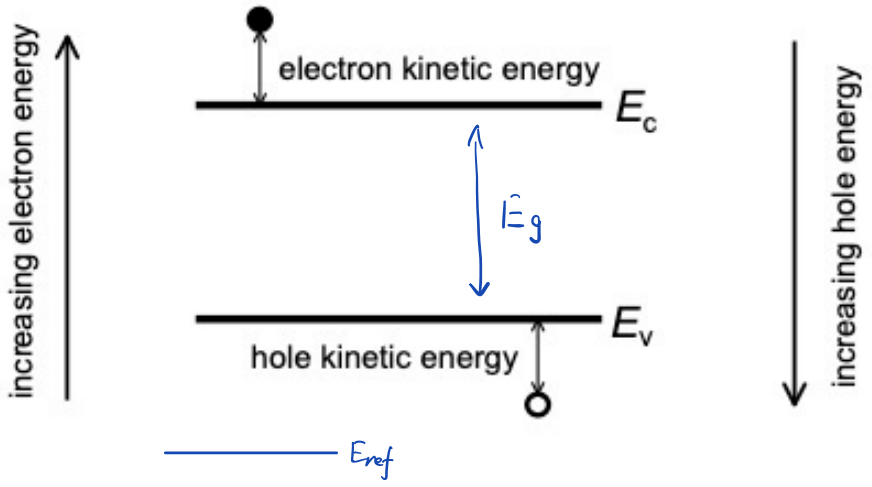


Total Current Density

- $J = J_{n|dft} + J_{n|diff} + J_{p|dft} + J_{p|diff}$
- $J = en\mu_n E + eD_n \frac{dn}{dx} + ep\mu_p E - eD_p \frac{dp}{dx}$
- This expression contains four terms
- Fortunately, in most situations, we only need to consider one term at any one time at a particular point in a semiconductor.



Band Diagram: Potential vs. Kinetic Energy



$E_c(x) = E_{ref} + (-e)\cdot V(x)$

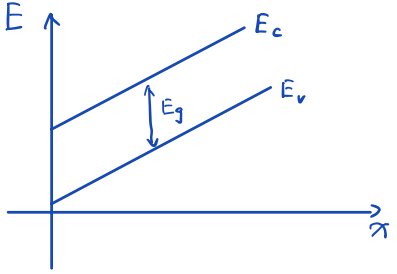
$E_v(x) = E_{ref} + (-e)\cdot V(x) - E_g$

$V(x) = -\int \mathcal{E}(x)\,dx = -\mathcal{E}\cdot x$

$\mathcal{E}(x) = \frac{dV(x)}{dx}$

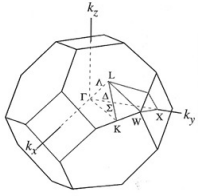
$E_c(x) = E_{ref} + e\cdot\mathcal{E}\cdot x$

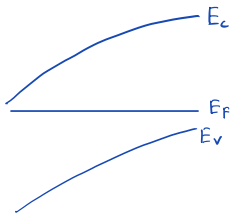
$E_v(x) = E_{ref} + e\cdot\mathcal{E}\cdot x - E_g$



E_c represents the electron potential energy:

P.E. = $E_c - E_{reference}$





$$N_d(x) = n(x) = N_c \cdot \exp\left(-\frac{E_c - E_F}{kT}\right) = n_i \cdot \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

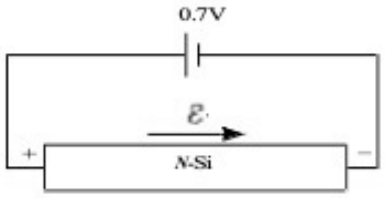
$$J_n = e \cdot n \cdot \mu_n \cdot E + e \cdot D_n \cdot \frac{\partial n}{\partial x} = 0$$

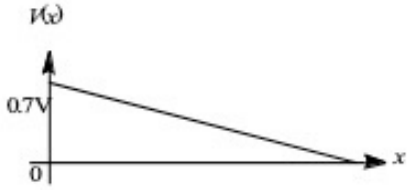
$$n = n_i \cdot \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

$$\Rightarrow J_n = e \cdot n \cdot \mu_n \cdot E + e \cdot D_n \cdot (-n) \cdot (eE) \cdot \frac{1}{kT} = 0$$

$$\frac{D_n}{\mu_n} = \frac{kT}{e}$$

Electrostatic Potential V

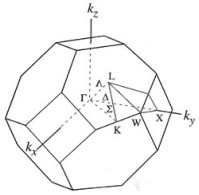




Potential energy of a –q charged particle is related to the electrostatic potential V(x):

$$\text{P.E.} = -qV$$

$V = (E_{reference} - E_c) / q$



Electrostatic Potential V

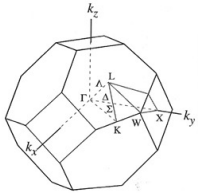


Potential energy of a –q charged particle is related to the electrostatic potential V(x):

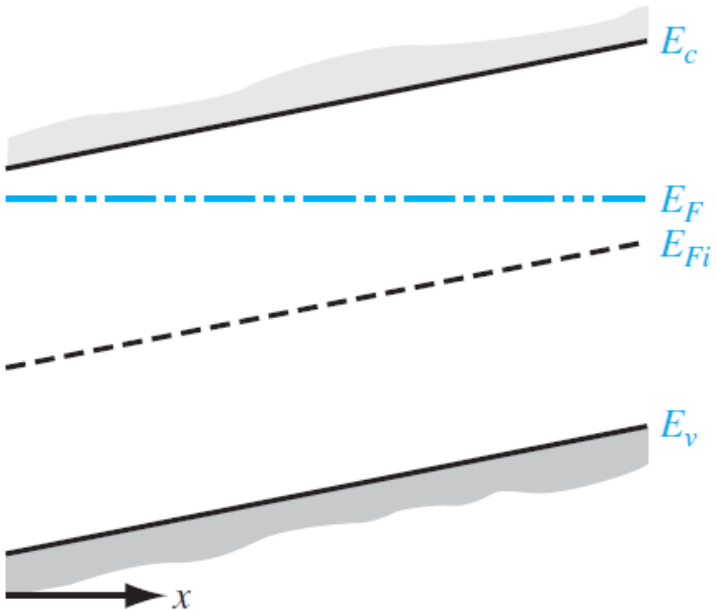
$$P.E. = -qV$$

$V = (E_{reference} - E_c) / q$

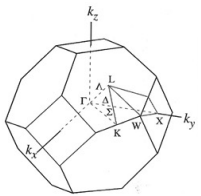
Variation of E_c with position is called “band bending.”



Graded Impurity Distribution

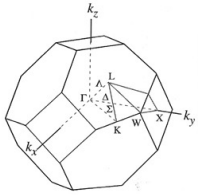


Question: What is the induced electric field?



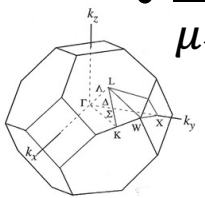
Induced Electric Field

- $E_x = -\frac{d\varphi}{dx} = \frac{1}{e}\frac{dE_{Fi}}{dx}$
- $n_0 = n_i \exp\left[\frac{(E_F - E_{Fi})}{kT}\right] \approx N_d(x)$
- $E_F - E_{Fi} = kT ln(\frac{N_d(x)}{n_i})$
- $-\frac{dE_{Fi}}{dx} = \frac{kT}{N_d(x)}\frac{dN_d(x)}{dx}$
- $E_x = -\frac{kT}{e}\frac{1}{N_d(x)}\frac{dN_d(x)}{dx}$



Einstein Relationship

- Consider the non-uniformly doped semiconductor in thermal equilibrium:
- $J_n = 0 = en\mu_n E_x + eD_n \frac{dn}{dx}$
- $J_n = 0 = eN_d(x)\mu_n E_x + eD_n \frac{dN_d(x)}{dx}$
- $E_x = -\frac{kT}{e}\frac{1}{N_d(x)}\frac{dN_d(x)}{dx}$
- $0 = eN_d(x)\mu_n \frac{-kT}{e}\frac{1}{N_d(x)}\frac{dN_d(x)}{dx} + eD_n \frac{dN_d(x)}{dx}$
- $\frac{D_n}{\mu_n} = \frac{kT}{e}$

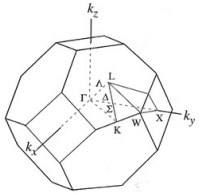


Einstein Relationship

- $\frac{D_n}{\mu_n} = \frac{kT}{e}$
- $\frac{D_p}{\mu_p} = \frac{kT}{e}$
- What is $\frac{kT}{e}$?

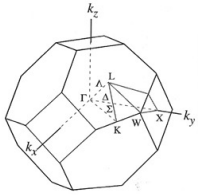
Table 5.2 | Typical mobility and diffusion coefficient values at $T = 300$ K (μ = cm²/V-s and D = cm²/s)

	μ_m	D_n	μ_p	D_p
Silicon	1350	35	480	12.4
Gallium arsenide	8500	220	400	10.4
Germanium	3900	101	1900	49.2



Hall Effect

- To distinguish whether a semiconductor is n type or p type
- To Measure the majority carrier concentration
- To measure majority carrier mobility



Hall Effect

Lorentz force $F = qv \times B$

After balance $F = q(E + v \times B) = 0$

$$qE_y = qv_xB_z$$

$V_H = \pm E_H W$ Depend on p type or n type

If p type:

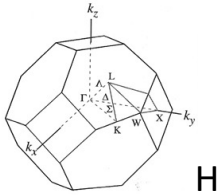
$$v_{dx} = \frac{J_x}{ep} = \frac{I_x}{(ep)(Wd)}$$

$$V_H = \frac{I_xB_z}{(epd)}$$

$$p = \frac{I_xB_z}{edV_H}$$ → Here we got carrier density

$$\frac{I_x}{(Wd)} = J_x = ep\mu_pE_x = ep\mu_p\frac{V_x}{L}$$

Here we got carrier mobility



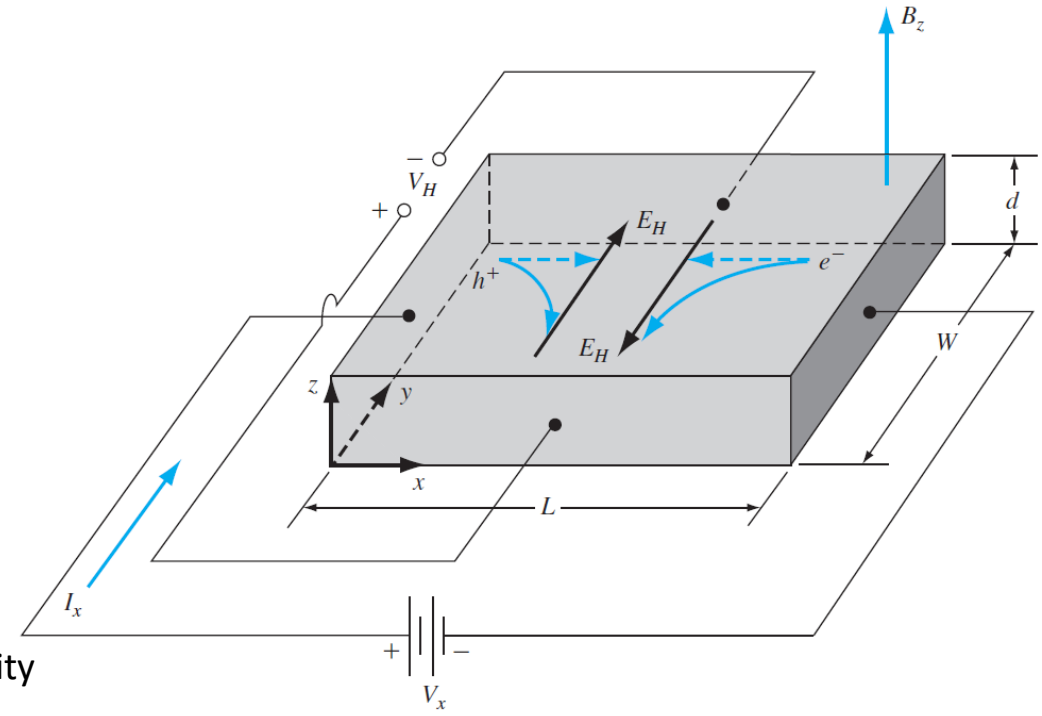
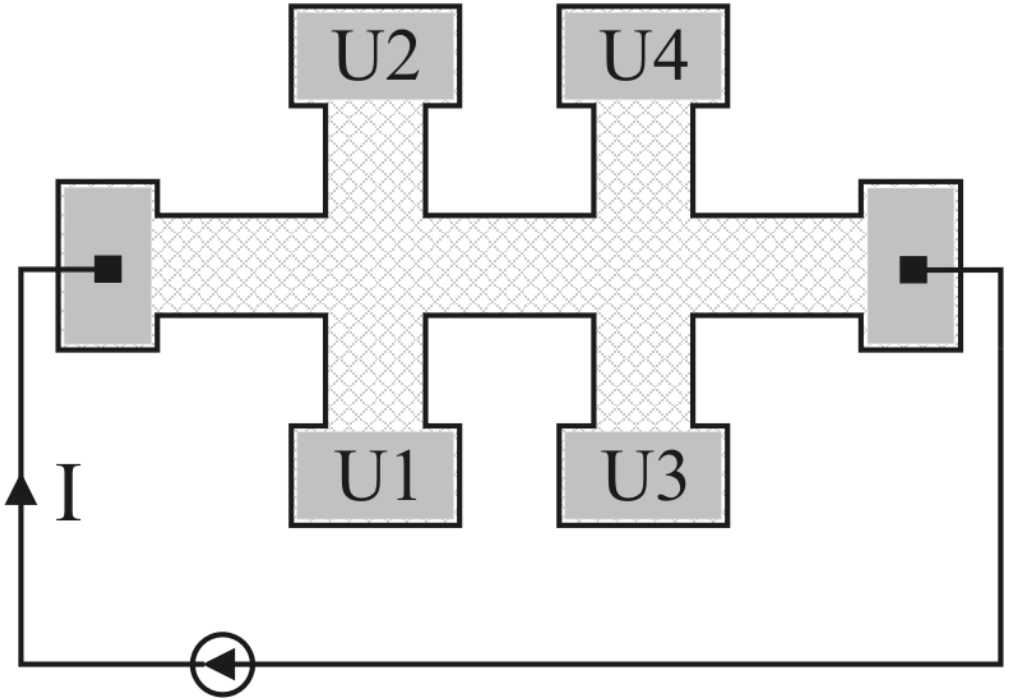
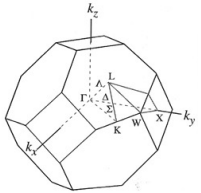


Figure 5.13 | Geometry for measuring the Hall effect.

Bridge Type Hall Measurement





"Semiconductor Material and Device Characterization" by Dieter Schroder